

2 1 C S S 3 0 3 T · D A T A S C I E N C E

Master Study Guide

by Aswath

[First Principles](#) · [PYQ Analysis](#) · [Easy Preparation](#) · [Memory Palace](#)

[May 2025](#) · [July 2025](#) · [December 2025](#) | [Max Marks: 75](#) | [Python](#)

What's Inside & How to Use

SECTION	WHAT IT IS	WHEN TO USE
PYQ Analysis	Part B probability table — 18 questions from 3 papers	Read first.
Easy Preparation	Most likely questions with exact exam answers	Study this if you have 1 day.
Master Checklist	Every topic with MCQ/B/C frequency	Tick off as you revise.
Memory Palace	Visual scenes for each unit	Walk through on exam day.
Unit 1–5	First principles — why each concept exists	Read once slowly.
MCQ Bank	40 questions + WHY the answer is correct	Cover and answer out loud.
Part B Answers	Model answers with key code	Read structure, not word-for-word.
Part C Answers	15-mark answers in three parts	Read once. Understand the flow.
Cheatsheet	Everything essential on one page	Last thing before exam.

STUDY METHOD — 3 PASSES

Pass 1: Read Units 1–5. Understand WHY, not just WHAT.

Pass 2: Easy Preparation. Cover the answer. Say it out loud. Check.

Pass 3: Walk the Memory Palace. Place every concept in the building.

Night before: Cheatsheet only. Nothing new.

Introduction — Read This First

What this guide is

This is a complete exam preparation guide for 21CSS303T Data Science at SRM IST. Built by analysing 3 official PYQ papers to find exactly which topics repeat, which questions are guaranteed, and what an examiner-approved answer looks like. Everything inside is evidence-based. Nothing is guessed.

Evidence — Papers Used

#	PAPER	SEMESTER	BATCH	MAX MARKS
1	21CSS303T — May 2025	Sixth Semester	2021–2025	75
2	21CSS303T — July 2025	Sixth Semester	2021–2025	75
3	21CSS303T — December 2025	Sixth Semester	2021–2025	75
4	21CSS303T Syllabus	SRM Reg. 2021	Common Course	—

Techniques Used in This Guide

FIRST PRINCIPLES

Instead of memorising a definition, you first understand WHY the concept was invented. The definition then becomes obvious. Example: You learn WHY NumPy exists — Python lists are slow because each element is a separate object. NumPy forces one type in one memory block. Now you understand the definition AND can answer any variation of the question. How to use it: Look for the FIRST PRINCIPLE box at the top of every unit.

MEMORY PALACE

An ancient technique where you imagine a familiar building. Each room holds one topic. When you need to recall something in the exam, you mentally walk through the building. Example: The GATE has a python snake with numbers falling out of its mouth — that is NumPy. Inside the LAUNDRY ROOM, janitors clean dirty clothes — that is data wrangling. Your brain stores images and locations 10x more reliably than abstract text. How to use it: Read the Memory Palace section once. Visualise each room. On exam day, close your eyes for 10 seconds and walk through.

MEMORY HOOK

A one-sentence trick that links a concept to something you already know. Not a full explanation — just an anchor. Example: DROP = dropna (row disappears). FILL = fillna (blank gets a value). You do not need to think — you just see the word DROP and you know which function. How to use it: After reading an answer, say the Memory Hook out loud once.

EXAM-READY FORMAT

All model answers are written exactly as you should write them in the exam. No academic padding. The original question from the actual PYQ paper is shown first — exactly as it appeared in print. This activates photographic memory: your brain recognises the pattern when you see the real question in the exam hall. How to use it: Read the original question first. Try to recall the answer. Then check. Repeat until you can recall without looking.

Model Questions — Exact Text from Official PYQ Papers

Copied directly from official exam papers. Read them exactly as you would in the exam hall.

B.Tech/M.Tech (Integrated) Degree Examination — May 2025

21CSS303T DATA SCIENCE · Sixth Semester · Max Marks: 75 · Time: 3 hours

PART B (4 × 10 = 40 MARKS) — ANSWER ANY 4

- | | |
|--------------------|--|
| Q21
10mk | Discuss the key steps involved in Data Science Process with neat sketch. |
| Q22
10mk | Analyse the various techniques to handle large datasets and list the challenges faced during data wrangling. |
| Q23
10mk | Develop a Python code to clean and transform data using Pandas functions like dropna(), fillna(), and replace(). |
| Q24
10mk | Discuss briefly about line plots, scatter plots, box plot and histogram used in data visualisation. |
| Q25
10mk | Categorise the steps involved in merging and concatenating datasets using Pandas. |
| Q26
10mk | Examine how to handle missing data and common techniques to deal with missing values. |

PART C (1 × 15 = 15 MARKS) — ANSWER ANY 1

- | | |
|--------------------|--|
| Q27
15mk | Explain various techniques used in data visualisation and develop a Python program to create a 3D surface plot using Matplotlib. |
| Q28
15mk | Discuss in detail about the data cleaning and wrangling techniques on a sample dataset of your choice using Pandas. |

B.Tech/M.Tech (Integrated) Degree Examination — July 2025

21CSS303T DATA SCIENCE · Sixth Semester · Max Marks: 75 · Time: 3 hours

PART B (4 × 10 = 40 MARKS) — ANSWER ANY 4

- | | |
|--------------------|---|
| Q21
10mk | (a) Write Python code to calculate the mean, maximum, and minimum temperature using NumPy. (5 Marks) (b) Write Python code to sort the temperatures in ascending order and find the index of the highest temperature. (5 Marks) |
|--------------------|---|

```
import numpy as np
temps = np.array([30, 32, 28, 35, 31, 29, 34])
```

Q22 10mk	(a) What is Web Scraping? Explain its applications and ethical concerns. (5 Marks) (b) Compare Web APIs and Open Data Sources as methods of data acquisition in Data Science. (5 Marks)
Q23 10mk	(a) Write Python code to create a scatter plot to visualise the relationship between house size and price. (5 Marks) (b) Use Seaborn to fit a regression line to the scatter plot. (5 Marks) <pre>import pandas as pd data = {'Size (sq ft)': [1000,1500,2000,2500,3000], 'Price (\$)': [150000,200000,250000,300000,350000]} df = pd.DataFrame(data)</pre>
Q24 10mk	(a) Describe a step-by-step approach to identify and handle duplicate records and missing values. (5 Marks) (b) Explain data standardisation and discuss two techniques to normalise data. (5 Marks)
Q25 10mk	(a) What are the different types of data visualisations? Explain any three with examples. (5 Marks) (b) Discuss how Matplotlib and Seaborn can be used for customising plots. Mention three key techniques. (5 Marks)
Q26 10mk	Using Seaborn, write Python code to: (a) Create a scatter plot between Marketing Spend and Sales Revenue. (5 Marks) (b) Use a regression line to show the trend between the two variables. (5 Marks)

P A R T C (1 × 15 = 15 M A R K S) — A N S W E R A N Y 1

Q27 15mk	a) Discuss the challenges faced when handling large datasets. Provide three general techniques to efficiently manage large volumes of data. (5 marks) b) Explain data wrangling operations: Merging datasets on an index, Concatenating datasets, Reshaping data using pivoting. (5 marks) c) Write Python code using Pandas to fill missing Salary values with the median salary per department, and identify and remove outliers in the Age column using the IQR method. (5 marks)
Q28 15mk	You are a data scientist analysing monthly sales data for an e-commerce company (sales revenue, customers, product categories, discounts over 3 years). a) Explain the role of different plot types in understanding sales data — line plot, scatter plot, histogram. (5 marks) b) Create a figure with four subplots using Matplotlib to display monthly sales trends per product category. (5 marks) c) Discuss three techniques for customising plots using Matplotlib and Seaborn. (5 marks)

B.Tech/M.Tech (Integrated) Degree Examination — December 2025

21CSS303T DATA SCIENCE · Sixth Semester · Max Marks: 75 · Time: 3 hours

P A R T B (4 × 10 = 40 M A R K S) — A N S W E R A N Y 4

Q21 10mk	A multinational retail company collects daily transactional data from hundreds of stores. Explain how NumPy arrays and Pandas DataFrames can be applied to manage and manipulate this data. Include details of array creation, indexing, slicing, joining, and aggregation.
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Q22 10mk	A healthcare organisation maintains electronic medical records with missing values in critical fields such as age, inconsistent categorical entries in diagnosis, and extreme outliers in test scores. Develop a structured data wrangling and cleaning process to address these issues.
Q23 10mk	A financial analyst preparing a quarterly report requires visualisations highlighting growth trends, regional comparisons, and anomalies. Evaluate the effectiveness of line plots, bar charts, histograms, boxplots, and pair plots using Matplotlib and Seaborn.
Q24 10mk	An e-commerce company aims to build a predictive model classifying whether a customer will purchase a product. Apply a supervised learning approach — data preparation, model selection (logistic regression / decision trees), train/test split — and evaluate using accuracy, precision, recall, and F1-score.
Q25 10mk	A telecommunications company seeks to segment its customer base using call duration, internet usage, recharge frequency, and complaint history. Analyse how K-means clustering identifies customer segments. Discuss preprocessing, number of clusters, and silhouette score.
Q26 10mk	A social media platform gathers millions of comments containing emojis, spelling variations, and irrelevant symbols. Create a complete text-preparation workflow: cleaning, tokenisation, stopword removal, normalisation, and transformation. Justify why each stage is essential.

P A R T C (1 × 15 = 15 M A R K S) — A N S W E R A N Y 1

Q27 15mk	A government health department collected COVID-19 patient records with missing values in age/gender, misspelled city names, inconsistent test results ('Positive','POS','+ve'), and age outliers (values above 200). Identify the data quality issues and explain why cleaning is necessary. (5 Marks) Write Python syntax to: Load dataset, Handle missing values, Standardise inconsistent entries, Remove age outliers. (10 Marks)
Q28 15mk	An insurance company has customer records with demographic details, premium paid, claims made, and settlement history. Build a predictive model classifying whether a new policyholder will make a claim. Dataset includes categorical and numerical variables requiring preprocessing. Evaluate suitability of Logistic Regression vs Decision Trees. (5 Marks) Write Python syntax to: Import and preprocess data, Split train/test, Train Decision Tree, Compute accuracy, precision, and recall. (10 Marks)

PYQ Analysis — What Actually Comes in the Exam

Based on all 3 papers: May 2025, July 2025, December 2025. 18 Part B slots analysed.

Part B Topic Probability

TOPIC	PAPERS	PROBABILITY	APPEARED IN
Data cleaning & missing values	3/3	100% ★★★	May Q23,Q26 · Jul Q24 · Dec Q22
Visualisation (charts + customise)	3/3	100% ★★★	May Q24 · Jul Q25,Q26 · Dec Q23
NumPy array operations	2/3	67% ★★	Jul Q21 · Dec Q21
Scatter plot + regression line	2/3	67% ★★	Jul Q23, Q26 (appeared TWICE in one paper)
Large dataset handling	2/3	67% ★★	May Q22 · Dec Q21
Web scraping & APIs	1/3	33% ★	Jul Q22
Pandas merge & concatenation	1/3	33% ★	May Q25
Data Science process	1/3	33% ★	May Q21
ML classification pipeline	1/3	33% ★	Dec Q24 — NEW, watch for this
K-means clustering	1/3	33% ★	Dec Q25 — NEW, watch for this
Text data pipeline	1/3	33% ★	Dec Q26 — NEW, watch for this

All 18 Part B Questions — Full Log

PAPER	Q	QUESTION ASKED	CO
May 2025	Q21	Data Science process — key steps with sketch	CO1
May 2025	Q22	Large datasets — handling techniques + challenges	CO2
May 2025	Q23	Data cleaning code — dropna(), fillna(), replace()	CO3
May 2025	Q24	Line, scatter, box plots, histogram	CO4
May 2025	Q25	Merging and concatenating datasets using Pandas	CO3
May 2025	Q26	Missing data — identification and handling	CO3
Jul 2025	Q21	NumPy temperature array — mean, max, min, sort, argmax	CO3
Jul 2025	Q22	Web scraping — applications + ethics + APIs	CO2
Jul 2025	Q23	House prices scatter plot + Seaborn regression line	CO3

Jul 2025	Q24	Duplicate + missing values + standardisation	CO2
Jul 2025	Q25	Visualisation types + Matplotlib/Seaborn customisation	CO4
Jul 2025	Q26	Seaborn scatter + regression: Marketing vs Sales	CO4
Dec 2025	Q21	Retail data — NumPy + Pandas: index, slice, join, aggregate	CO1
Dec 2025	Q22	Healthcare records — data wrangling + cleaning pipeline	CO2
Dec 2025	Q23	Financial report — multiple visualisation techniques	CO3
Dec 2025	Q24	E-commerce ML — classification + evaluation metrics	CO4
Dec 2025	Q25	Telecom — K-means clustering + silhouette score	CO5
Dec 2025	Q26	Social media text — cleaning + tokenisation pipeline	CO2

KEY INSIGHT

Data cleaning + Visualisation appeared in 100% of papers. Know both cold.
Scatter + regression appeared TWICE in July 2025 alone. Very high signal.
Dec 2025 introduced ML and clustering. These are likely to repeat.
Strategy: prepare the top 5 topics. You answer only 4. You always have a choice.

Easy Preparation — Most Likely Questions & Exact Answers

If you have only 1 day — study this section only.

Every question is selected from 100% or 67% probability topics.

Answers are written exactly as you should write them in the exam hall.

MCQ Preparation — 3 Most Probable Question Types

These exact patterns appeared in every paper. Know these = 6 guaranteed marks in Part A minimum.

MCQ Type 1 | ★★★ 100% — every paper

Missing Value Handling

Which function is used to remove missing values in Pandas?

A	✓ dropna()
B	fillna()
C	replace()
D	remove_nan()
WHY	There are only two choices when data is missing: remove the row or fill the hole. dropna() removes entire rows containing NaN. fillna() fills the blank with a value and keeps the row.
MEMORY	DROP = dropna (row disappears). FILL = fillna (blank gets a value). Two words. Two actions.
EXAM TIP	Any MCQ asking about 'remove NaN' or 'handle missing data' — dropna removes, fillna fills. Options like 'remove_nan' or 'clear_nan' do not exist in Pandas — eliminate them immediately.

MCQ Type 2 | ★★★ 100% — every paper

NumPy Identity Matrix & Array Ops

Which function in NumPy creates an identity matrix?

A	np.zeros()
B	np.ones()
C	np.eye()
D	✓ np.diag()
WHY	An identity matrix has 1s on the diagonal and 0s everywhere else. np.eye(n) and np.diag([1,1,...,1]) both create it. diag() is the generalised diagonal form. If options show both, pick diag.
MEMORY	eye() = I = Identity. np.eye(3) = 3x3 with 1s on diagonal. diag = diagonal. argmax = INDEX not value.

EXAM TIP

Three NumPy questions rotate: identity matrix = eye/diag. Array of zeros = zeros(). Index of max = argmax(). Knowing which does what is worth 3 marks across all papers.

MCQ Type 3 | ★★★ 100% — every paper

Box Plot vs Histogram Purpose

A box plot is primarily used to ____.

A	Show trends over time
B	Display frequency of categories
C	✓ Show distribution and detect outliers
D	Visualise correlation between two variables
WHY	Box = IQR (Q1 to Q3 = middle 50% spread). Whiskers extend 1.5×IQR beyond the box. Any point outside whiskers = outlier. Two jobs: distribution (box) + outlier detection (dots beyond whiskers).
MEMORY	Box = IQR. Whiskers = 1.5×IQR fence. Dots = outliers. Histogram = distribution of 1 variable. Line = over time.
EXAM TIP	Trick: 'over time' = line chart. 'Frequency of categories' = bar chart. 'Distribution one variable' = histogram. 'Spread + outliers' = box plot. These four appear as distractors for each other.

Part B Preparation — Top 4 Questions Ranked by Probability

Answer any 4 from 6. Prepare Rank 1 and 2 cold = 20 secured marks. Rank 3 or 4 = safety net.

Part B — Rank 1 ★★★★★ 100% — ALL 3 papers

Data cleaning: missing values, outliers & standardisation

Appeared in: May 2025 Q23 + Q26 · Jul 2025 Q24 · Dec 2025 Q22

FIRST PRINCIPLE	All real-world data is dirty. Missing = form skipped. Outlier = typo (age 250 instead of 25). A model trained on dirty data learns the noise, not the signal. Garbage in = garbage out.
MEMORY	Laundry room. dropna = throw torn items. fillna(median) = patch holes. IQR fence = remove oversized clothes. Min-Max machine = tag everything with a 0-1 number.
EXAM ANSWER	· Define: Data cleaning is the process of detecting and fixing errors, missing values, outliers, and inconsistencies before analysis. · Missing values: two choices — dropna() if < 5% rows missing and missing is random. fillna(median) for numerical columns, fillna(mode) for categorical. · Outliers — IQR method: $IQR = Q3 - Q1$. Lower fence = $Q1 - 1.5 \times IQR$. Upper fence = $Q3 + 1.5 \times IQR$. Remove rows outside the fence. · Standardisation: Min-Max scales to $[0,1]$ using $(x - \min) / (\max - \min)$. Z-score scales to mean=0, std=1 using $(x - \text{mean}) / \text{std}$. Purpose: prevent large-scale features dominating the model.
KEY CODE	<pre>df.isnull().sum() # count NaN per column df['Age'].fillna(df['Age'].median(), inplace=True) # fill numeric NaN Q1, Q3 = df['Age'].quantile(.25), df['Age'].quantile(.75) IQR = Q3 - Q1 df = df[(df['Age'] >= Q1 - 1.5 * IQR) & (df['Age'] <= Q3 + 1.5 * IQR)] # remove outliers</pre>
CONCLUDE	Data cleaning is the foundation of every reliable analysis. No algorithm can correct bad training data. Each step — filling NaN, removing outliers, standardising scales — directly increases model accuracy.

Part B — Rank 2 ★★★★★ 100% — ALL 3 papers

Visualisation: chart types + Seaborn customisation

Appeared in: May 2025 Q24 · Jul 2025 Q25 + Q26 · Dec 2025 Q23

FIRST PRINCIPLE	Numbers do not show patterns to humans directly. The brain processes shapes in milliseconds. Visualisation converts data into shapes. Right chart = right question answered. Matplotlib is the engine. Seaborn is the statistical layer on top.
MEMORY	Art studio. Four paintings — mountain = histogram, box with whiskers = box plot, stars = scatter, river = line chart. Seaborn = the curator who adds regression lines and clean themes.
EXAM ANSWER	· Histogram: distribution of ONE numerical variable. Bins count frequency. plt.hist(df['Age'], bins=20). When to use: How is this variable spread? · Box plot: spread + outliers. Box = $Q1$ to $Q3$. Whiskers = $1.5 \times IQR$. Dots beyond = outliers. sns.boxplot(y=df['Salary']).

	· Scatter + regression (regplot): relationship between two variables + best-fit line. <code>sns.regplot(x='X',y='Y',data=df)</code>. · Customise: <code>plt.title()</code> + <code>xlabel()</code> + <code>ylabel()</code> for labels. <code>plt.subplots(r,c)</code> for multiple charts. <code>plt.savefig('f.png')</code> BEFORE <code>show()</code>. <code>sns.set_theme()</code> for style.
KEY CODE	<pre>plt.hist(df['Age'], bins=20) # histogram sns.boxplot(y=df['Salary']) # box plot sns.regplot(x='X', y='Y', data=df) # scatter + regression line fig,ax=plt.subplots(1,2,figsize=(10,4)) # side-by-side charts plt.tight_layout() plt.savefig('out.png') # layout + save</pre>
CONCLUDE	Visualisation turns data into decisions. Choosing the right chart for the right question is a core data science skill. Seaborn's regplot is the most efficient tool for two related variables.

Part B — Rank 3 ★★ 67% — 2 of 3 papers (twice in Jul 2025)

Scatter plot + regression line using Seaborn

Appeared in: Jul 2025 Q23 + Q26 · May 2025 Q24 (partial)

FIRST PRINCIPLE	Scatter shows where individual points sit — does X correlate with Y? The regression line shows the overall direction — the best-fit straight line minimising squared errors. <code>sns.regplot()</code> does both in one call.
MEMORY	Stars in the sky = scatter (individual points). River connecting all stars = regression line (trend). <code>sns.regplot</code> = both in one line.
EXAM ANSWER	· Define: A scatter plot visualises the relationship between two numerical variables. Each dot = one data point. · The regression line is the best-fit straight line through all points. Positive slope = positive correlation. Negative slope = inverse. Flat = no relationship. · <code>sns.regplot()</code> plots scatter AND regression in one call. The shaded band = 95% confidence interval. · Always label axes. Call <code>plt.savefig('f.png')</code> BEFORE <code>plt.show()</code> — <code>show()</code> clears the figure.
KEY CODE	<pre>sns.regplot(x='Marketing_Spend', y='Sales', data=df) plt.title('Marketing vs Sales') plt.xlabel('Spend') plt.ylabel('Revenue') plt.savefig('trend.png') # save BEFORE show()</pre>
CONCLUDE	The scatter + regression combination is the standard tool for understanding two related variables. A positive slope tells a business that more marketing drives more sales — that is actionable, quantified insight.

Part B — Rank 4 ★★ 67% — 2 of 3 papers

NumPy array operations on a dataset

Appeared in: Jul 2025 Q21 · Dec 2025 Q21

FIRST PRINCIPLE	Python lists are slow because every element is a separate Python object. NumPy stores all elements as the same C-type in one memory block. One instruction operates on all elements simultaneously — a vectorized operation. 100× faster than a Python loop.
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MEMORY	NumPy = army marching together (one command, all act). Python list loop = one soldier at a time. <code>eye()</code> = I = Identity. <code>argmax</code> = INDEX not value.
EXAM ANSWER	· Define: NumPy is a Python library for fast, vectorized numerical operations on arrays stored in contiguous memory. · <code>np.array([30,32,28,35])</code> — creates 1D array. <code>np.mean()</code>, <code>np.max()</code>, <code>np.min()</code> — statistics on whole array. · <code>np.argmax(arr)</code> — returns the INDEX of the maximum value, not the value itself. Index is 0-based. · <code>np.sort(arr)</code> — returns a NEW sorted copy. Does NOT change the original array. · <code>np.eye(3)</code> — 3×3 identity matrix (1s on diagonal, 0s elsewhere).
KEY CODE	<pre>arr = np.array([30, 32, 28, 35, 31]) np.mean(arr) np.max(arr) np.min(arr) # 31.2, 35, 28 np.argmax(arr) # index 3 (position of 35) np.sort(arr) # [28 30 31 32 35] — new array np.eye(3) # 3x3 identity matrix</pre>
CONCLUDE	NumPy's vectorized operations eliminate slow Python loops. <code>argmax</code> vs <code>max</code> , and <code>sort</code> returning a copy — these two details are the most commonly tested in exams.

Part C Preparation — Top 2 Questions Ranked by Probability

Answer any 1 question. 15 marks = three parts of 5 marks each. 1.5 pages is enough. Theory + code + conclusion.

Part C — Rank 1 15 marks — answer any 1

Data cleaning pipeline — identify issues + clean a real dataset

Appeared in: Dec 2025 Q27 · Dec 2025 Q28 · Cleaning pattern in ALL 3 papers

FIRST PRINCIPLE	Every real dataset has exactly four types of problems: blanks (missing), typos (outliers), copy-paste (duplicates), inconsistency (same value written differently). Each has a specific Pandas fix. A model trained on dirty data learns the noise — making cleaning non-negotiable.
MEMORY	Four types of dirt, four Pandas fixes. Missing = fillna(median). Inconsistent = map(). Misspelled = str.title(). Outlier = IQR fence.
(A) 5 MARKS — THEORY	· Identify four data quality issues: (1) Missing values in age/gender — causes models to learn with incomplete, biased data. (2) Inconsistent labels — 'Positive','POS','+ve' treated as 3 different classes — wrong distribution. (3) Outliers — age 250 is a typo for 25 — skews mean and distorts normalisation. (4) Misspellings — 'mumbai','Mumbai','Mumbay' = one city split into three groups. · State the Pandas fix for each: fillna(median) for numeric NaN, fillna(mode) for categorical, map() for label standardisation, str.strip().str.title() for spelling, IQR method for outliers.
(B) 5 MARKS — CODE	<pre>df['Age'].fillna(df['Age'].median(), inplace=True) # fill missing age df['City'] = df['City'].str.strip().str.title() # fix misspellings m={'Positive':'Positive','POS':'Positive','+ve':'Positive'} df['Result'] = df['Result'].map(m) # standardise labels Q1,Q3=df['Age'].quantile(.25),df['Age'].quantile(.75) IQR=Q3-Q1 df=df[(df['Age']>=Q1-1.5*IQR) & (df['Age']<=Q3+1.5*IQR)] # remove outliers</pre>
(C) 5 MARKS — CONCLUSION	· Data cleaning ensures the dataset is accurate (correct values), consistent (uniform format), and complete (no blanks). · Used median not mean for imputation because median is robust to the same outliers being cleaned. · str.strip().str.title() fixes both capitalisation and leading/trailing spaces in one call. · IQR adapts to the actual data distribution — better than an arbitrary cutoff. Garbage in = garbage out — cleaning prevents this.

Part C — Rank 2 15 marks — answer any 1

NumPy + Pandas operations on a large business dataset

Appeared in: Dec 2025 Q21 · May 2025 Q22 · Jul 2025 Q21 (overlap)

FIRST PRINCIPLE	Python alone cannot handle millions of rows at usable speed. NumPy stores same-type elements in contiguous memory — C-speed computation. Pandas adds labelled indexing and relational operations on top. NumPy = engine. Pandas = steering wheel. Both are needed.
MEMORY	NumPy = engine (maths on numbers fast). Pandas = steering wheel (find, filter, join data by label).
(A) 5 MARKS — THEORY	· NumPy's role: stores data as same-type elements in contiguous memory. Vectorized operations (sum, mean, sort) run in C-speed on the whole array at once. Never write a Python loop on NumPy data. · Pandas' role: wraps NumPy with a labelled index. Enables SQL-style operations — filter rows, group by category, merge two tables on a shared column, reshape with pivot tables. · For a retail dataset: <code>pd.read_csv()</code> to load, <code>head()</code> and <code>describe()</code> to inspect, <code>groupby('City')['Sales'].sum()</code> to aggregate, <code>merge(products, on='ProductID')</code> to join, <code>pivot_table()</code> to reshape.
(B) 5 MARKS — CODE	<pre>df = pd.read_csv('retail.csv') # load df.head() df.describe() df.isnull().sum() # inspect df.groupby('City')['Sales'].sum() # aggregate by city merged = df.merge(products, on='ProductID') # join two tables arr = np.array([5200,4800,6100]) # NumPy array np.mean(arr) np.argmax(arr) arr[1:3] # stats + slice</pre>
(C) 5 MARKS — CONCLUSION	· NumPy's vectorized operations eliminate Python loops, making computations 100× faster on large datasets. · Pandas' labelled indexing makes code readable — you query by meaning (<code>df['City']</code>) not by position. · Together they form the complete data pipeline — NumPy handles computation, Pandas handles structure. This is the prerequisite for all downstream ML models.

Master Study Checklist

Tick each topic as you revise. ★ = exam frequency. Use this the night before.

TOPIC	MCQ	PART B	PART C	KEY THING TO REMEMBER
NumPy — array, eye, zeros, argmax	★★★	★★	—	argmax = INDEX not value. eye(3) = identity. sort = new copy.
Pandas — Series vs DataFrame	★★★	★	—	Series = 1D labelled. DataFrame = 2D table.
df.merge, dropna, fillna, head	★★★	★★	★	merge(on='col'). dropna = remove. fillna(median) = fill.
Data cleaning — missing + outliers	★★★	★★★★	★★★★	IQR = Q3-Q1. Fence = $\pm 1.5 \times \text{IQR}$. fillna(median) for nums.
Standardisation	★★	★★	★	Min-Max $\rightarrow [0,1]$. Z-score $\rightarrow \text{mean}=0, \text{std}=1$.
One-hot encoding	★★	★	—	Text categories $\rightarrow$ binary 0/1 columns for ML.
Web scraping	★	★	—	BeautifulSoup. Automated HTML extraction. Check robots.txt.
Histogram	★★★	★	—	1 variable, distribution, adjust bins for detail.
Box plot	★★★	★	—	Box = IQR. Whiskers = $1.5 \times \text{IQR}$. Dots = outliers.
Scatter + regression	★★	★★★★	★	sns.regplot(x,y,data). savefig BEFORE show().
Subplots + savefig	★	★	★	plt.subplots(r,c). savefig BEFORE show().
Seaborn pairplot	★★	★	—	sns.pairplot(df) = all pairwise scatter plots.
Supervised vs unsupervised	★★	—	★	Supervised = has labels. Unsupervised = finds patterns.
Linear vs Logistic Regression	★★	★	★	Linear = number output. Logistic = category output.
Train/test split + overfitting	★★	★	★	80/20. Overfit = high train, low test accuracy.
Precision + Recall + F1	★★	★	★	Precision = $\text{TP}/(\text{TP}+\text{FP})$. Recall = $\text{TP}/(\text{TP}+\text{FN})$.
K-means clustering	★★	★	★	K chairs. Assign nearest. Move to mean. Repeat.
Silhouette score + PCA	★	★	★	-1 to 1. Closer to 1 = well-separated. PCA = reduce dims.

Exam Pattern — Quick Reference

PART	QUESTIONS	PER Q	TOTAL	RULE
Part A	20 MCQs	1 mark	20 marks	Answer ALL — no choice
Part B	6 Questions	10 marks	40 marks	Answer ANY 4
Part C	2 Questions	15 marks	15 marks	Answer ANY 1

EXAM TIME PLAN

0–20 min: Part A. All 20 MCQs. Eliminate 2 wrong options first if unsure.

20–90 min: Part B. Read all 6 first. Then write your best 4.

90–130 min: Part C. Pick 1. Theory (5) + code (5) + conclusion (5). 1.5 pages.

Key insight: Part B + C = 55/75 marks. Most time goes here.

Memory Palace — The Data Science Building

Walk through this building on exam day. Each room = one unit. If you can place the concept, you can write about it.

Gate — NumPy & Pandas

A giant PYTHON SNAKE with NUMBERS falling from its mouth = NumPy (vectorized, C-speed). Behind it: PANDA BEARS reading books with labelled index tabs = Pandas (labelled tables). This is your entry to every exam question.

Laundry Room — Data Wrangling

Janitors clean dirty clothes = dirty data. dropna = throw torn items. fillna(median) = patch holes with average fabric. IQR fence = sort items too big or small (outliers). One-hot machine = sew number tags on all items so ML can read them.

Art Studio — Visualisation

Four paintings on the wall: mountain silhouette = histogram (distribution). Box with whiskers = box plot (spread + outliers). Stars scattered = scatter plot (two variables). River flowing = line chart (over time). Seaborn is the curator adding regression lines and colour themes.

Laboratory — Machine Learning

Scientists run experiments. 80% of samples go to training. 20% locked away for testing (unseen data). A student who memorised only last year's exact exam answers fails new questions = overfitting.

Boardroom — Clustering

Manager places K chairs randomly (K centroids). Everyone sits at the nearest chair (assignment). Manager moves each chair to the centre of its group (update). Repeat until no one moves (convergence). Silhouette score = how clearly separated the groups are.

Unit 1 — Data Science, NumPy & Pandas

FIRST PRINCIPLE — WHY DATA SCIENCE EXISTS

Every decision made without data relies on guessing. DS exists because guessing at scale is expensive.

Pipeline: Raw Data → Clean → Analyse → Insight → Decision.

Three roles = three jobs: Engineer (build data pipes), Scientist (find patterns), ML Engineer (ship models).

DATA TYPES

- Structured: tables with fixed schema. SQL, Excel. Easy to query.
- Semi-structured: tags/keys but no fixed shape. JSON, XML.
- Unstructured: no format. Images, video, raw text.

NumPy

FIRST PRINCIPLE

Python lists = each element a separate Python object = slow. NumPy = same C-type, contiguous memory = one instruction on ALL elements = 100× faster. Never loop over a NumPy array. Use `np.mean()`, `np.sum()` etc.

```
arr = np.array([30,32,28,35])  np.mean(arr)  np.max(arr)  np.min(arr)
np.argmax(arr)  # INDEX of max      np.sort(arr)  # new sorted copy
np.eye(3)      # identity matrix  np.zeros((3,3)) np.concatenate([a,b])
```

MEMORY HOOK

`argmax` = INDEX not value. `eye` = I = Identity. `sort` = returns copy, not in-place.

Pandas

FIRST PRINCIPLE

NumPy uses 0,1,2 as indices. Real data needs 'Date','Name','City'. Pandas adds labelled index. Series = 1D. DataFrame = 2D table.

```
df=pd.read_csv('f')  df.head()  df.describe()  df.isnull().sum()
df.dropna()  df.fillna(df.mean())  df.drop_duplicates()
df.sort_values('col')
df.merge(df2, on='col')  df.reset_index()
```

MEMORY HOOK

`dropna` = DROP rows. `fillna` = FILL blanks. `merge` = SQL JOIN. `describe` = all stats in 1 call.

Unit 2 — Data Wrangling & Cleaning

FIRST PRINCIPLE

Real data is always dirty. Missing = form skipped. Outlier = typo. Duplicate = merged databases. Garbage in = garbage out. No algorithm can fix bad training data.

MISSING DATA

- `dropna()` — use when < 5% missing AND missing at random. Loses rows.
- `fillna(median)` — numerical column with outliers. Median is outlier-robust.
- `fillna(mode)` — categorical column. Most common category.

```
df['Age'].fillna(df['Age'].median())
df['City'].fillna(df['City'].mode()[0])
```

OUTLIER DETECTION — IQR

FIRST PRINCIPLE

IQR = $Q3 - Q1$ = middle 50% spread. Fence = $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$. Points outside = outliers.

```
Q1,Q3=df['col'].quantile(.25),df['col'].quantile(.75); IQR=Q3-Q1
df=df[(df['col']>=Q1-1.5*IQR)&(df['col']<=Q3+1.5*IQR)]
```

SCALING

- Min-Max: $(x - \min) / (\max - \min) \rightarrow [0, 1]$. Use for KNN, neural networks.
- Z-score: $(x - \text{mean}) / \text{std} \rightarrow \text{mean}=0, \text{std}=1$. Use for regression, SVM.

```
df['n']=(df['col']-df['col'].min())/(df['col'].max()-df['col'].min()) #
Min-Max
df['s']=(df['col']-df['col'].mean())/df['col'].std() # Z-
score
```

Unit 3 — Visualisation

FIRST PRINCIPLE

The brain processes shapes in milliseconds. Visualisation converts data into shapes humans can read instantly. Matplotlib = drawing engine. Seaborn = statistical charts built on top.

CHART	USE WHEN	CODE
Histogram	Distribution of 1 numerical variable	<code>plt.hist(df['Age'], bins=20)</code>
Box plot	Spread + outliers of 1 variable	<code>sns.boxplot(y=df['col'])</code>
Scatter	Relationship between 2 variables	<code>sns.scatterplot(x='X',y='Y',data=df)</code>
Regplot	Scatter + regression line in 1 call	<code>sns.regplot(x='X',y='Y',data=df)</code>
Line chart	Trend over time	<code>plt.plot(x, y)</code>
Pair plot	All pairwise relationships	<code>sns.pairplot(df)</code>

```
plt.title('T') plt.xlabel('X') plt.ylabel('Y') # always label
plt.savefig('f.png') # BEFORE show()
fig,ax=plt.subplots(r,c) plt.tight_layout() # multiple charts
```

Unit 4 — Machine Learning (Supervised)

FIRST PRINCIPLE

Supervised = learning with a teacher (labels). Number output → Regression. Category output → Classification.

- Linear Regression: best-fit line. Output: continuous number.
- Logistic Regression: output probability 0–1, threshold at 0.5. Output: YES/NO.
- Overfitting: memorised noise. High train acc + low test acc.

```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2)
model=LogisticRegression(); model.fit(X_train,y_train);
model.score(X_test,y_test)
```

METRIC	FORMULA	PLAIN MEANING
Accuracy	$(TP+TN)/\text{Total}$	% of all predictions correct
Precision	$TP/(TP+FP)$	When I say YES — how often right?
Recall	$TP/(TP+FN)$	Of all actual YES — how many caught?
F1 Score	$2 \times (P \times R) / (P + R)$	Balance of Precision and Recall

MEMORY HOOK

Precision = don't cry wolf. Recall = don't miss any fire. F1 = balance both.
TP = said YES was YES. FP = said YES was NO. FN = said NO was YES.

Unit 5 — Unsupervised Learning & Clustering

FIRST PRINCIPLE

Unsupervised = no teacher, no labels. Find hidden structure in raw data. K-means: group similar points by minimising intra-cluster distance.

- Step 1: Choose K (number of clusters — your only decision).
- Step 2: Place K centroids randomly.
- Step 3: Assign each point to its NEAREST centroid.
- Step 4: Move each centroid to the MEAN of its assigned points.
- Step 5: Repeat until nothing moves (convergence).
- Silhouette Score: -1 to 1. Closer to 1 = well-separated groups.
- PCA: reduces many dimensions into fewer principal components while keeping maximum variance.

```
km=KMeans(n_clusters=3,random_state=42); km.fit(X); labels=km.labels_
```

MEMORY HOOK

K chairs. Everyone sits at nearest chair. Move chair to centre of group. Repeat until stable.
Wrong K? Dissimilar people on same chair = poor clusters.

MCQ Bank — 40 Questions with Explanation

Cover the blue tick row. Say your answer out loud. Then reveal. If wrong — re-read WHY.

Set 1 — July 2025

Q1. Primary goal of Data Science?	
A	Store large amounts of data
B	✓ Analyze and extract insights from data
C	Design websites
D	Create databases
W H Y	DS exists to convert data into decisions. Storage is IT infrastructure — not DS.
H O O K	DS = Data → Insight → Decision.
Q2. Which operation is faster in NumPy vs Python lists?	
A	Looping element by element
B	String manipulations
C	Defining functions
D	✓ Vectorized operations
W H Y	NumPy runs in compiled C on whole array at once. Python loops interpret bytecode one step at a time.
H O O K	NumPy = army marching together. Python loop = soldiers one by one.
Q3. Function to create a NumPy array?	
A	✓ array()
B	create_array()
C	list_to_numpy()
D	make_array()
W H Y	np.array() is the standard constructor. Others do not exist in NumPy.
H O O K	np.array([1,2,3]) — just wrap your list.
Q4. Difference: Pandas Series vs DataFrame?	
A	Series holds multiple columns
B	✓ Series is one-dimensional, DataFrame is two-dimensional
C	DataFrame is immutable
D	Series does not support indexing
W H Y	Series = one labelled column (1D). DataFrame = multiple Series sharing an index (2D table).
H O O K	Series = 1 shelf. DataFrame = whole library.
Q5. Function to remove missing values in Pandas?	

A	✓ dropna()
B	fillna()
C	replace()
D	remove_nan()
W H Y	dropna() removes rows with NaN. fillna() fills them.
H O O K	DROP = dropna. FILL = fillna.
Q6. Main challenge with large datasets?	
A	Storing in Excel
B	✓ High computational cost and memory usage
C	Creating charts
D	Adding rows manually
W H Y	Large data doesn't fit in RAM. Requires chunking, distributed computing.
H O O K	Big data = big compute cost.
Q7. Pandas function to merge DataFrames on a common column?	
A	✓ merge()
B	concat()
C	join()
D	combine()
W H Y	merge() = SQL JOIN on shared column. concat() stacks. join() uses index.
H O O K	merge(on='col') = SQL JOIN.
Q8. Technique to detect outliers?	
A	Min-Max Scaling
B	Standardization
C	✓ Interquartile Range (IQR)
D	Label Encoding
W H Y	IQR sets fence $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$. Points outside = outliers.
H O O K	IQR = fence. Outside = outlier.
Q9. Purpose of Web Scraping?	
A	Manually download data
B	Improve SEO rankings
C	Delete unwanted data
D	✓ Extract structured data from web pages automatically
W H Y	Scraping is AUTOMATED extraction from HTML. Not manual.
H O O K	Scraping = automated data harvest from HTML.
Q10. Function to reset index of a DataFrame?	
A	✓ reset_index()
B	reindex()

C	set_index() reindex()
D	reset_index() and set_index()
W H Y	reset_index() resets to default 0,1,2. set_index('col') sets a column as the new index.
H O O K	Reset = back to default 0,1,2.

Q11. Most common visualisation library in Python?

A	NumPy
B	✓ Matplotlib
C	Pandas
D	Scikit-learn
W H Y	Matplotlib is the foundation. Seaborn builds on it.
H O O K	Matplotlib = canvas and paint of Python charts.

Q12. Best plot for distribution of one numerical variable?

A	Bar Plot
B	Scatter Plot
C	✓ Histogram
D	Pie Chart
W H Y	Histogram bins continuous data and counts frequency.
H O O K	Histogram = mountain range shape = one variable distribution.

Q13. Purpose of a Box Plot?

A	✓ Display spread and detect outliers
B	Show trends over time
C	Visualise categorical variables
D	Compare two datasets
W H Y	Box = IQR. Whiskers = $1.5 \times \text{IQR}$. Dots outside = outliers.
H O O K	Box = IQR. Whiskers = fence. Dots = outliers.

Q14. Function to read a CSV file in Pandas?

A	pd.open_csv()
B	✓ pd.read_csv()
C	pd.load_csv()
D	pd.get_csv()
W H Y	pd.read_csv() is the correct standard function.
H O O K	Read CSV = pd.read_csv(). Nothing more.

Q15. Best chart for trends over time?

A	Bar Chart
B	Pie Chart
C	✓ Line Chart
D	Scatter Plot

W H Y	Line connects points in time order showing trends.
H O O K	Line = river flowing through time.
Q16. Python library for web scraping?	
A	NumPy
B	Pandas
C	✓ BeautifulSoup
D	Scikit-learn
W H Y	BeautifulSoup parses HTML and extracts data.
H O O K	BeautifulSoup = parsing the HTML soup.
Q17. Function to sort a Pandas DataFrame by column?	
A	df.sort()
B	df.orderby()
C	df.arrange()
D	✓ df.sort_values()
W H Y	df.sort_values('col') is correct. Others do not exist in Pandas.
H O O K	sort_values = sort by column values.
Q18. Function to check first few rows of DataFrame?	
A	df.top()
B	✓ df.head()
C	df.first()
D	df.show()
W H Y	df.head(5) returns first 5 rows by default.
H O O K	head() = head of document = first rows.
Q19. Function to remove duplicate rows?	
A	✓ drop_duplicates()
B	remove_duplicates()
C	clear_duplicates()
D	delete_duplicates()
W H Y	drop_duplicates() is the valid Pandas method.
H O O K	DROP the DUPLICATES. Literal name.
Q20. Function to create array of zeros in NumPy?	
A	np.empty()
B	np.ones()
C	✓ np.zeros()
D	np.full()
W H Y	np.zeros(shape) fills with 0.0. np.ones() fills with 1.0.

H O O K

`zeros()` = zero heroes. All zeroes.

Set 2 — December 2025

Q1. NumPy function to create an identity matrix?	
A	eye()
B	ones()
C	zeros()
D	✓ diag()
W H Y	Both np.eye(n) and np.diag([1,...,1]) create identity matrices. diag() is the generalised diagonal form.
H O O K	eye() = I = Identity. diag = diagonal of 1s.
Q2. Pandas structure for labeled one-dimensional data?	
A	✓ Series
B	DataFrame
C	Panel
D	Matrix
W H Y	Series = 1D array plus a labelled index. DataFrame = 2D.
H O O K	Series = one series of data. One shelf.
Q3. Method to remove missing values from a DataFrame?	
A	fillna()
B	✓ dropna()
C	replace()
D	reindex()
W H Y	dropna() removes NaN rows. fillna() replaces them.
H O O K	DROP = dropna. FILL = fillna.
Q4. Which step is NOT part of the Data Science process?	
A	Data collection
B	Data visualization
C	Model evaluation
D	✓ Software installation
W H Y	Software installation is pre-requisite setup, not a DS pipeline step.
H O O K	Install software = IT setup. Not data science.
Q5. Method to merge DataFrames on a common column?	
A	concat()
B	append()
C	✓ merge()
D	join()
W H Y	merge() = SQL JOIN on column. concat() stacks. join() uses index.

H O O K	merge() = SQL JOIN.
Q6. Why handle missing data?	
A	Reduces dataset size
B	✓ Prevents model bias and errors
C	Speeds up computation
D	Increases redundancy
W H Y	NaN causes models to crash or learn biased patterns.
H O O K	NaN = poison for models.
Q7. Which function detects outliers?	
A	mean()
B	std()
C	✓ boxplot()
D	reindex()
W H Y	A boxplot visually shows outliers as dots beyond whiskers ($1.5 \times \text{IQR}$).
H O O K	boxplot = built-in outlier visualiser.
Q8. Best transformation for categorical text data (for ML)?	
A	Normalization
B	✓ One-hot encoding
C	Standardization
D	Missing value imputation
W H Y	ML needs numbers. One-hot converts 'Red' $\rightarrow$ [1,0,0], 'Blue' $\rightarrow$ [0,1,0].
H O O K	One column per category. 1 where it matches, 0 elsewhere.
Q9. Which library provides pairplot()?	
A	matplotlib
B	✓ seaborn
C	pandas
D	plotly
W H Y	seaborn.pairplot() = all pairwise scatter plots.
H O O K	Seaborn = pairplot is its signature feature.
Q10. A boxplot is primarily used to ____.	
A	Display frequency of categories
B	✓ Show distribution and detect outliers
C	Plot linear relationships
D	Visualize correlation
W H Y	Box = IQR. Whiskers = $1.5 \times \text{IQR}$. Dots beyond = outliers.
H O O K	Box = distribution. Dots = outliers.

Q11. Which matplotlib command saves plot to file?

A plt.show()

B ✓ plt.savefig()

C plt.plot()

D plt.write()

W H Y plt.savefig() saves to disk. plt.show() only displays on screen.

H O O K savefig = save figure. Call BEFORE show().

Q12. Histogram bins too wide — which parameter fixes this?

A linewidth

B alpha

C ✓ bins

D figsize

W H Y bins controls how many intervals data is split into. More bins = finer detail.

H O O K More bins = thinner slices = finer detail.

Q13. Which is a supervised learning algorithm?

A K-means

B ✓ Linear Regression

C PCA

D Apriori

W H Y Supervised needs labels. Linear Regression learns from labelled data to predict a number.

H O O K Supervised = teacher gives answers (labels).

Q14. Purpose of train/test split?

A Reduce computation

B ✓ Evaluate on unseen data

C Increase dataset size

D Avoid missing values

W H Y Training and testing on same data = cheating. Test set = real-world simulation.

H O O K Test set = exam for the model. New questions only.

Q15. Best algorithm when output is a category?

A Linear Regression

B ✓ Logistic Regression

C K-means

D PCA

W H Y Logistic Regression outputs probability 0–1 for classification.

H O O K Category output = Logistic Regression.

Q16. High train accuracy + low test accuracy means ____.

A	Underfitting
B	✓ Overfitting
C	Normal fitting
D	Good generalisation
W H Y	Model memorised training data including noise. Cannot generalise.
H O O K	Overfit = memorised examples, fails new questions.
Q17. Which algorithm is used for clustering?	
A	Linear Regression
B	Logistic Regression
C	✓ K-means
D	Decision Tree
W H Y	K-means is the standard unsupervised clustering algorithm.
H O O K	Clustering without labels = K-means.
Q18. Primary goal of unsupervised learning?	
A	Predict outcomes
B	✓ Find hidden patterns in data
C	Classify into labeled groups
D	Evaluate supervised models
W H Y	Unsupervised = no labels. Goal is to discover hidden structure.
H O O K	Unsupervised = explorer. No map. Find the pattern.
Q19. Method for dimensionality reduction?	
A	KNN
B	✓ PCA
C	Random Forest
D	Naive Bayes
W H Y	PCA transforms high-dimensional data into fewer components while retaining maximum variance.
H O O K	PCA = compress 10 dimensions into 2 key slides.
Q20. Dissimilar points in same cluster — what does this mean?	
A	Low variance in data
B	✓ Poor choice of number of clusters
C	Correct clustering
D	Overfitting
W H Y	If K is too small, dissimilar points are forced into the same cluster.
H O O K	Dissimilar in same group = K is too small.

Part B — Model Answers (10 marks each)

Structure: definition + first principle + steps/explanation + key code (3–5 lines) + conclusion.

RULE FOR CODE IN PART B

Write only the key lines — 3 to 5 is enough. Add a comment on each line explaining what it does. Full marks come from understanding, not from writing a full program.

B1 — NumPy array operations (Jul Q21)

`temps = np.array([30, 32, 28, 35, 31, 29, 34])` — calculate mean/max/min, sort and find index of highest.

NumPy exists because Python loops are too slow. Vectorized operations work on the whole array in C-speed — one instruction, all elements. Use built-in functions, never a for loop.

```
arr = np.array([30, 32, 28, 35, 31, 29, 34])
np.mean(arr)      # 31.28      np.max(arr)   # 35      np.min(arr)   # 28
np.sort(arr)      # [28 29 30 31 32 34 35] — returns NEW array
np.argmax(arr)    # 3 — index of 35 in the ORIGINAL array (0-based)
```

- `argmax()` = INDEX of the maximum, not the value itself. Very commonly tested.
- `np.sort()` returns a copy — original array is unchanged.

B2 — Data cleaning: duplicates + missing + standardisation (Jul Q24)

PART (A)

- Step 1: `df.duplicated().sum()` counts duplicates. `df.isnull().sum()` counts NaN per column.
- Step 2: `df.drop_duplicates(inplace=True)`.
- Step 3: `fillna(median)` for numerical, `fillna(mode)` for categorical.

```
df.duplicated().sum()          # count duplicates
df.drop_duplicates(inplace=True) # remove them
df['Age'].fillna(df['Age'].median(), inplace=True) # fill numeric NaN
```

PART (B)

First principle: features have different scales — age 0–100 vs salary 0–100,000. Without scaling, salary dominates every model. Standardisation puts all features on the same playing field.

- Min-Max: $(x - \min) / (\max - \min) \rightarrow [0, 1]$. Use for KNN, neural networks.
- Z-score: $(x - \text{mean}) / \text{std} \rightarrow \text{mean}=0, \text{std}=1$. Use for regression, SVM.

```
df['Age_norm'] = (df['Age'] - df['Age'].min()) / (df['Age'].max() - df['Age'].min())
df['Age_std'] = (df['Age'] - df['Age'].mean()) / df['Age'].std()
```

B3 — Scatter plot + Seaborn regression line (Jul Q23, Q26)

Scatter reveals whether X and Y are correlated. Regression line shows the overall direction. `sns.regplot()` does both in one call.

```
sns.regplot(x='Marketing_Spend', y='Sales', data=df)
plt.title('Marketing vs Sales') plt.xlabel('Spend') plt.ylabel('Revenue')
plt.savefig('trend.png') # save BEFORE show()
```

- Positive slope = more X → more Y (positive correlation).
- Shaded band = 95% confidence interval. Wider = more uncertainty.
- `savefig` must come BEFORE `show()` — calling `show()` clears the figure.

B4 — Visualisation types + customisation (Jul Q25)

PART (A)

- Histogram: distribution of ONE numerical variable. `plt.hist(df['Age'], bins=20)`. Use when: how is this variable spread?
- Box Plot: spread + outliers. Box=Q1-Q3. Whiskers=1.5×IQR. Dots=outliers. `sns.boxplot(y=df['Salary'])`.
- Scatter: relationship between TWO variables. `sns.scatterplot(x='X', y='Y', data=df)`.

PART (B)

- Labels and title: `plt.title()`, `plt.xlabel()`, `plt.ylabel()`. Makes chart self-explanatory.
- Figure size and subplots: `plt.figure(figsize=(10,6))`. `plt.subplots(2,2)`. `plt.tight_layout()`.
- Seaborn themes: `sns.set_theme(style='whitegrid')`. `sns.set_palette('viridis')`.

```
fig, ax = plt.subplots(1, 2, figsize=(12,5)) # 2 charts
ax[0].hist(df['Age'], bins=15) # histogram left
sns.boxplot(y=df['Salary'], ax=ax[1]) # box plot right
plt.tight_layout() plt.savefig('charts.png') # layout + save
```

Part C — Model Answers (15 marks each)

Structure: (a) theory + first principles [5 marks] + (b) key Python code [5 marks] + (c) analysis + conclusion [5 marks]. 1.5 pages total.

C1 — Data cleaning pipeline (Dec 2025 Q27 pattern)

Govt COVID-19 records: missing age/gender, inconsistent test results, age outliers, misspelled cities.

(A) IDENTIFY ISSUES — 5 MARKS

- Issue 1 — Missing values: age and gender blank. Models learn biased patterns from incomplete data.
- Issue 2 — Inconsistent labels: 'Positive', 'POS', '+ve' = same class but treated as 3 different categories. Wrong class distribution.
- Issue 3 — Outliers: age 250 is a typo for 25. One bad value shifts mean and distorts normalisation.
- Issue 4 — Misspellings: 'mumbai', 'Mumbai', 'Mumbay' = one city split into 3 groups. Wrong aggregation.

(B) CLEANING CODE — 5 MARKS

```
df['Age'].fillna(df['Age'].median(), inplace=True) # fill missing age
df['Gender'].fillna(df['Gender'].mode()[0], inplace=True) # fill missing gender
m={'Positive':'Positive','POS':'Positive','+ve':'Positive'}
df['Result']=df['Result'].map(m) # standardise labels
df['City']=df['City'].str.strip().str.title() # fix spelling
Q1,Q3=df['Age'].quantile(.25),df['Age'].quantile(.75); IQR=Q3-Q1
df=df[(df['Age']>=Q1-1.5*IQR)&(df['Age']<=Q3+1.5*IQR)] # remove outliers
```

(C) WHY THIS MATTERS — 5 MARKS

- Used median (not mean) for imputation because median is robust to outliers in the same column.
- `str.strip().str.title()` fixes both spaces and capitalisation in one line.
- IQR adapts to actual data distribution — better than an arbitrary cutoff like 'age > 120'.
- Result: accurate, consistent, complete data. Garbage in = garbage out — cleaning prevents this.

C2 — NumPy + Pandas on large retail dataset (Dec 2025 Q21)

(A) THEORY — 5 MARKS

- First principle: Python alone cannot handle millions of rows at usable speed. NumPy provides C-speed computation. Pandas adds labelled, queryable tables.

- NumPy role: numerical computation — sums, means, slicing, matrix operations. Never write a loop.
- Pandas role: filter rows, group by category, join two tables on shared column, reshape.
- Challenge with large data: memory overflow. Solution: `pd.read_csv(chunksize=10000)`.

(B) KEY CODE — 5 MARKS

```
df=pd.read_csv('retail.csv')           # load
df.head() df.describe() df.isnull().sum() # inspect
df.groupby('City')['Sales'].sum()       # aggregate by city
merged=df.merge(products, on='ProductID') # join two tables
arr=np.array([5200,4800,6100])          # NumPy array
np.mean(arr) np.argmax(arr) arr[0:2]    # stats + slicing
```

(C) CONCLUSION — 5 MARKS

- NumPy eliminates Python loops — 100× faster computation on large numerical arrays.
- Pandas' labelled indexing makes code readable — you query by meaning, not by position.
- Together they form the complete data pipeline — NumPy handles computation, Pandas handles structure. This is the prerequisite for all ML models.

Exam Day Cheatsheet — Read This Last

One page. Everything essential. Read only this the night before. Nothing new.

FUNCTION	WHAT IT DOES	FREQUENCY
<code>df.isnull().sum() + df.dropna() + df.fillna(median)</code>	Detect → remove → fill missing values	★★★★
<code>Q1,Q3=quantile(.25,.75) IQR=Q3-Q1</code> filter by fence	IQR outlier detection and removal	★★★★
<code>df.merge(df2,on='col')</code> <code>df.drop_duplicates()</code>	Join tables + remove duplicates	★★★
<code>df.head()</code> <code>df.describe()</code> <code>df.sort_values('col')</code>	Inspect + sort data	★★★
<code>np.array()</code> <code>np.mean()</code> <code>np.argmax()</code> <code>np.sort()</code>	NumPy core operations	★★★
<code>np.eye(3)</code> <code>np.zeros((r,c))</code> <code>np.concatenate([a,b])</code>	Identity + zeros + join arrays	★★
<code>plt.hist(df['X'],bins=20)</code> <code>sns.boxplot(y=df['Y'])</code>	Histogram + box plot	★★★
<code>sns.regplot(x='X',y='Y',data=df)</code>	Scatter + regression line in 1 call	★★★
<code>fig,ax=plt.subplots(r,c)</code> <code>plt.tight_layout()</code> <code>plt.savefig</code>	Multiple charts + save	★★★
<code>train_test_split(X,y,test_size=0.2)</code> <code>model.fit()</code> <code>score()</code>	Train + test ML model	★★
<code>KMeans(n_clusters=3).fit(X)</code> <code>km.labels_</code>	K-means clustering	★★

Memory Palace — walk this before entering the exam hall

GATE: NumPy snake (numbers) + Pandas bears (labelled books)

LAUNDRY: dropna, fillna(median), IQR fence, one-hot machine

ART STUDIO: mountain (hist), whiskers (box), stars (scatter), river (line)

LAB: train/test, overfitting = memorised noise, Logistic = YES/NO

BOARDROOM: K chairs, assign nearest, move to mean, repeat, silhouette

FINAL EXAM STRATEGY

Part A: eliminate 2 wrong options first. Data cleaning + visualisation = 6 guaranteed marks.

Part B: read ALL 6 questions first. Pick Rank 1 (cleaning) + Rank 2 (visualisation) = 20 marks secured.

Part C: pick 1. Write (a) theory 5mk + (b) 4–5 lines code 5mk + (c) conclusion 5mk. 1.5 pages. savefig BEFORE show. `argmax` = INDEX not value. `dropna` = remove, `fillna` = fill.

